

AisleBot Network — Self-Hosted Pi Access Point

The Pi is now its own WiFi network. No router, no eduroam, no IP hunting. Power it on, it broadcasts `AisleBot-Pi`, you join, you work.

The fixed numbers

What	Value
Network name (SSID)	<code>AisleBot-Pi</code>
Password	<code>aislebotpi5</code>
Pi address (always)	<code>10.42.0.1</code>
Dashboard	<code>http://10.42.0.1:8080</code>
SSH	<code>ssh aritra@10.42.0.1</code>

10.42.0.1 never changes. NetworkManager's shared mode always puts the AP host at that address, so the dashboard and SSH targets are the same every single session. Your phone and PC get `10.42.0.x` handed out automatically by the Pi's own DHCP.

Daily use

1. Power the Pi. Give it about 30 seconds to boot and bring the radio up in AP mode.
2. On phone and/or PC, join `AisleBot-Pi` (password `aislebotpi5`).
3. Phone: browser to `http://10.42.0.1:8080` for the dashboard.
4. PC: `ssh aritra@10.42.0.1` for a terminal.

Phone and PC can both be on it at the same time. Drive from the phone, watch logs over SSH on the PC, no conflict.

Not done yet — make it survive reboot (Phase 2)

Right now the AP does **not** start automatically on boot. Autoconnect is off, left over from testing as a safety net. If you reboot the Pi today it goes back to eduroam, and 10.42.0.1 won't exist until you bring the AP up by hand.

This Pi runs networking through netplan (NetworkManager is just the renderer underneath), and its live profiles sit in `/run`, which is wiped on every reboot. So locking the AP in as the permanent default means writing it into netplan's config, not relying on the autoconnect flag. That's the next step.

Manual start, until Phase 2 is done:

```
bash
sudo nmcli con up aislebot-ap
```

When you need internet on the Pi (apt, git, pip)

The Pi has one radio. As its own AP it has no path to the internet. To go online for a task, switch it back to eduroam:

```
bash
sudo nmcli con up eduroam
```

Your PC then rejoins eduroam and reaches the Pi at its `10.53.x.x` address. Switch back home with:

```
bash
sudo nmcli con up aislebot-ap
```

Each switch drops your current SSH session, because the radio is changing networks. Reconnect on the new one. That's expected, not a fault.

What did NOT change

ROS2 runs exactly as before. CycloneDDS is bound to loopback, so node discovery never looked at WiFi in the first place. The dashboard code is unchanged; it binds to all interfaces and now just answers on 10.42.0.1. The ESP32 serial bridge, the udev rules, and `aislebot.service` are all untouched.

ESP32 WiFi backup — keep it

The ESP32's own AP stays. It is the escape hatch for when the Pi itself is dead: crash, SD corruption, kernel panic, USB drop. The PID lives on the ESP32, so even with the Pi gone you connect straight to it and drive the robot out of an aisle or stop it.

What	Value
SSID	AisleBot-Control
Password	aislebot123
Address	http://192.168.4.1

To use it, switch your phone off `AisleBot-Pi` and onto `AisleBot-Control`. You're on one network at a time, so reaching the backup is a deliberate "the Pi is down, switch networks" move.

If the dashboard ever fails to load but SSH works

First suspect is the firewall. Check:

```
bash
```

```
sudo ufw status
```

If it's active and blocking 8080, that's the fix to chase.